Date 14 / 10 / 26

Lec 28

Class and Object

Eg.1

```
class student:
    def __init__(self, name, age):
        print(id(self))
        self.name = name
        self.age = age
    def display_info(self):
        print("student name is", self.name)
        print("student age is", self.age)

s1 = student("rahul", 20)
print(s1.age)
print(id(s1))

s2 = student("Akshay", 25)
s2.display_info()
```

attribute

method

object reference s1, s2

objects

Eg2.

```
class school:
    def __init__(self, school_name, city):
        self.school_name = school_name
        self.school_city = school_city
    def display_info(self):
        print("ABC school")
        print("school name is:", self.school_name)
        print("school city is:", self.school_city)
```

Date __/__/____

```
school1 = School("ABC school", "Pune")
school2 = school("xyz school", "mumbai")

print(school1.school_name)
print(school1.~~city~~ school_city)

print(school2.school_name)
print(school2.school_city)

school1.display_info()
school2.display_info()
```

Method & function:

	Method	Function
1.	Define inside the class	Define outside the class
2.	call through object	Call directly
3.	call syntax: Obj_name.method_name()	call syntax: method_name()
4.	exists self	Does not exist self
5.	eg. list - append(), remove(), count(), insert(), copy()	Built in functions: len(), min(), max(), sum(), type(), id(), sorted(), reversed()

class list:
def append(self):
def remove(self):

class tuple:

class set:

6. def sum():
def len():

Page No.

Date __/__/____

Constructor:

--init-- method

def sum_calculate(): ← normal method
sum_calculate() → function call

constructor:
def ---init--(self):

Object1 = class1()

~ --init-- is the only method which executes automatically,

~ When we create an object of our class. This is superpower of constructor.

~ But we have to call other methods explictly.

	Normal method	--init--method Constructor
1.	explicitly call	When we create an obj automatically calls it.
2.	syntax: obj1.method_name()	Syntax: Obj1 = class_name()
3.	User has control	No user control

Date __/__/____

Constructor: We write configuration related code inside the constructor that we want to execute automatically without using action.

1] Database Connection code.
2] All Connection setup.

self: self is the reference of current object of the class.

There are two components in class.
1] attributes, properties, data
2] methods.

We can access it only through object
print(obj.attribute_name)
Obj.method_name()

We use self in 3places:

1] **Method Defination -**

```
Class Vehical:          <- Parameter
    def __init__(self):
```

Whenever we call method first parameter must be self.

2] **Attribute name -**

```
Class Vehical:
    def __init__(self, color, speed)
        self.color = color
        self.speed = speed
```

using self with Attribute name

Date __/__/____

3] Method calls inside class

```
class vehical:

    def __init__(self, speed):
        self.speed = speed

    def method1(self):
        ___
        ___

    def method2(self):
        ___
        ___
        self.method1()
```